

Sharp ordinal bounds for Hoare powerspaces

Extremal statures and the complete
 $\omega + n$ spectrum

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Abstract

The stature of a Noetherian space is the ordinal rank of the top element of its lattice of closed subsets. We study the stature of its extended Hoare powerspace, equipped with the lower Vietoris topology and including the empty closed set. Motivated by Question 1 in Section 15 of Goubault-Larrecq and Laboureix, we give a proof of the universal bound $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$, where exponentiation is ordinal. The binary upper bound is already known for the finitary powersets of well-quasi-orders; the argument here extends it to all Noetherian spaces. Explicit T_1 examples attain the upper bound at every ordinal stature. Consequently the older bound ω^α is attainable at infinite input stature α exactly when $\omega^\alpha = \alpha$.

We also determine the entire spectrum at each stature $\omega + n$, $n < \omega$. Its elements are precisely $\omega j(U) + j(Q) - j(U)$, where Q ranges over n -element posets, U ranges over their downsets, and $j(P)$ counts the downsets of P . The proof isolates a canonical infinite closed core, removes a possible generic point without changing the closed lattice, and computes a finite-cell rank. Every permitted value is realized by an explicit countable space. An accompanying program enumerates the finite spectra through $n = 6$ and checks a separate finite-model identity.

Status and scope. This is an original research attempt with full English proofs, not an independently refereed or proof-assistant-verified paper. The general spectrum beyond the finite tails of ω is not classified here. Priority is not asserted: the distinction between the known wqo theorem and the proposed general extension is explicit.

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1 The chosen question and the results

The problem selected for this investigation is a concrete ordinal-spectrum question rather than a foundational consistency question. It has two advantages for an attempted solution: the invariant is defined by a well-founded recursion, and the relevant hyperspace has a finite normal form for its closed sets.

Goubault-Larrecq and Laboureix establish

$$1 + \|X\| \leq \|\mathcal{H}_{0V}X\| \leq \omega^{\|X\|}. \quad (1.1)$$

Their Section 15, Question 1 asks which intermediate ordinals occur, and whether the upper endpoint can occur for *every* infinite value of $\|X\|$ [1, Section 15, Question 1]. We write HX for their $\mathcal{H}_{0V}X$ throughout. In particular, our powerspace includes the empty closed subset of X ; omitting that point would change several finite and successor formulas.

For an ordinal α , define

$$\text{Spec}_H(\alpha) = \{\|HX\| : X \text{ is Noetherian and } \|X\| = \alpha\}.$$

All constructions take place in ordinary set theory with choice. Spaces may have arbitrary cardinality and need not satisfy a separation axiom unless one is stated.

Main results proved in this manuscript

For every ordinal α ,

$$\min \text{Spec}_H(\alpha) = 1 + \alpha, \quad \max \text{Spec}_H(\alpha) = 2^\alpha.$$

The maximum has a T_1 witness. For infinite α ,

$$\omega^\alpha \in \text{Spec}_H(\alpha) \iff \omega \cdot \alpha = \alpha.$$

For $n < \omega$, with $j(P)$ the number of downsets of a finite poset P ,

$$\text{Spec}_H(\omega + n) = \{\omega \cdot j(U) + (j(Q) - j(U)) : |Q| = n, U \in \mathcal{J}(Q)\}.$$

Here U carries the induced order, and the empty downset is counted.

The simplest obstruction to universal upper attainment is already decisive: $\text{Spec}_H(\omega) = \{\omega\}$, so ω^ω is not attainable at input stature ω . The stronger result is not just this counterexample, but the optimal upper bound for every input and a criterion for exactly when the old bound coincides with it.

1.1 What is already known, and what is being extended

Abriola, Halfon, Lopez, Schmitz, Schnoebelen, and Vialard prove

$$1 + o(A) \leq o(\mathcal{P}_f(A)) \leq 2^{o(A)}$$

for well-quasi-orders, and provide sharpness witnesses for all ordinals [2, Theorems 3.3 and 4.3]. Thus neither binary ordinal exponentiation nor its sharpness in the wqo setting is a new discovery here. Our finite/limit/successor proof strategy is closely parallel to that existing theorem. The added step is a topological rank decomposition that works without an Alexandroff topology or a wqo specialization order.

This distinction matters. An infinite T_1 Noetherian space has equality as its specialization order, an infinite antichain, so it does not come from a wqo in this way. The cofinite topology is the first example. The T_1 witnesses below therefore give a realization in a genuinely different class of spaces, although unrestricted sharpness witnesses already follow from the known wqo results.

The proofs of the upper bound and the $\omega + n$ spectrum are supplied from rank and topology definitions. A separate known result, the product formula for statures, is explicitly imported only for the transfinite T_1 sharpness construction. The source question and these relevant primary papers were checked in their public versions; this is not a certification that no later or unpublished work contains the same extension.

2 Ranks, finite pieces, and closed–open decompositions

2.1 Definitions and elementary monotonicity

Definition 2.1. A space X is *Noetherian* when its closed subsets satisfy the descending chain condition. Equivalently, the inclusion order on $C(X)$, its set of closed subsets, is well-founded. For $F \in C(X)$, put

$$r_X(F) = \sup\{r_X(E) + 1 : E \in C(X), E \subsetneq F\}, \quad \|X\| = r_X(X).$$

The supremum of the empty family is 0.

Thus $\|\emptyset\| = 0$. If F is closed in X , its closed subsets are exactly the members of the interval $[\emptyset, F]$ in $C(X)$. Consequently

$$\|F\| = r_X(F). \quad (2.1)$$

There is no extra 1 in this identity: stature is the rank of the top point, not the rank of the whole closed-set poset including that top.

For well-founded posets, a strictly increasing map never decreases point ranks. Indeed, this follows by induction on the source point, since its strict predecessors map to strict predecessors of its image. We use this simple observation repeatedly.

Lemma 2.2. *If Y is a subspace of a Noetherian space X , then Y is Noetherian and $\|Y\| \leq \|X\|$. If $f : X \rightarrow Z$ is continuous and onto, then Z is Noetherian and $\|Z\| \leq \|X\|$.*

Proof. For the subspace assertion, send a closed $A \subseteq Y$ to $\text{cl}_X(A)$. The equality $Y \cap \text{cl}_X(A) = A$ shows that this is an order embedding of closed-set lattices. Its image is bounded by X , so the rank assertion follows. It also prohibits an infinite descending chain of closed subsets of Y .

For the image assertion, inverse image under f is an order embedding $C(Z) \hookrightarrow C(X)$, because f is onto. It sends Z to X . Apply the same reasoning. $\square$

The Kolmogorov, or T_0 , quotient identifies points with identical closures. Every closed subset is saturated for this equivalence, so the quotient has an isomorphic closed-set lattice. Passing to this quotient therefore preserves stature. Later we will see directly that it also preserves the homeomorphism type of the Hoare powerspace.

2.2 Natural sums and finite unions

The *natural sum* $\alpha \oplus \beta$ adds the coefficients in a common Cantor normal form. The *natural product* $\alpha \otimes \beta$ multiplies the corresponding polynomials in ω , using natural sums for exponents. These operations are commutative and are different from ordinary ordinal addition and multiplication.

Lemma 2.3. *For well-founded posets P, Q ,*

$$\text{rk}_{P \times Q}(p, q) = \text{rk}_P(p) \oplus \text{rk}_Q(q).$$

Consequently, for a finite topological disjoint sum,

$$\|X_1 \sqcup \cdots \sqcup X_k\| = \|X_1\| \oplus \cdots \oplus \|X_k\|.$$

Proof. Natural sum satisfies the recursion

$$\alpha \oplus \beta = \sup(\{(\alpha' \oplus \beta) + 1 : \alpha' < \alpha\} \cup \{(\alpha \oplus \beta') + 1 : \beta' < \beta\}).$$

The ranks realized strictly below a point of rank α include every ordinal below α : a proof appears in Lemma 2.5 below and is independent of this lemma. Applying the displayed recursion and induction on (p, q) gives the product formula. Predecessors that decrease both coordinates do not change the supremum, since they lie below a predecessor that decreases just one coordinate.

The lattice of closed subsets of a finite disjoint sum is the product of the closed-set lattices of its summands. Evaluate the formula at its top element and iterate. $\square$

Corollary 2.4. *For finitely many Noetherian subspaces Y_i whose union is Y ,*

$$\|Y\| \leq \bigoplus_i \|Y_i\|.$$

The Y_i need not be closed in Y .

Proof. The map from the finite topological disjoint sum of the Y_i onto their union is continuous. Use Lemmas 2.2 and 2.3. $\square$

Lemma 2.5 (No gaps in point ranks). *If p has rank α in a well-founded poset and $\beta < \alpha$, there is $q < p$ with rank exactly β .*

Proof. Choose a minimal member q of $\{x : x \leq p, \text{rk}(x) \geq \beta\}$. Every strict predecessor of q has rank less than β , so the rank recursion gives $\text{rk}(q) \leq \beta$. The reverse inequality holds by the choice of the set. Since $\beta < \text{rk}(p)$, $q \neq p$. $\square$

Lemma 2.6. *A finite T_0 space on m points has stature m . Conversely, if a T_0 space has finite stature m , it has exactly m points.*

Proof. The closed sets of a finite T_0 space are the downsets of its specialization poset. Strict inclusion increases cardinality, giving an upper bound m on the rank of the top. Initial segments of a linear extension give a chain of $m + 1$ closed sets from $\emptyset$ to X , so the rank is m .

An infinite T_0 space contains finite T_0 subspaces of every finite cardinality. By Lemma 2.2, its stature is at least every finite ordinal. This excludes finite stature for an infinite T_0 space. $\square$

2.3 The decomposition that drives the successor step

Lemma 2.7 (Closed–open rank inequality). *For a closed subset F of a Noetherian space X , put $V = X \setminus F$. Then*

$$\|F\| + \|V\| \leq \|X\| \leq \|F\| \oplus \|V\|. \quad (2.2)$$

The addition on the left is ordinary ordinal addition.

Proof. Closed subsets B of V correspond, by $B \mapsto F \cup B$, to the interval $[F, X]$ in $C(X)$. To verify this, write $B = V \cap C$ with C closed in X ; then $F \cup B = F \cup C$ is closed.

An induction on B gives

$$r_X(F \cup B) \geq \|F\| + r_V(B).$$

At $B = \emptyset$ this is equality. At a nonempty B , use predecessors $F \cup B'$ with $B' \subsetneq B$ and the continuity of ordinary addition in its right argument. Evaluating at $B = V$ proves the lower inequality.

For the upper inequality, the identity from the finer disjoint-sum topology $F \sqcup V$ onto X is continuous. Apply the finite-union bound. $\square$

Corollary 2.8 (Finite tail decomposition). *Let X be T_0 with $\|X\| = \lambda + n$, where λ is a nonzero limit ordinal and $0 < n < \omega$. There exists a closed $F \subseteq X$ with $\|F\| = \lambda$, and its complement has exactly n points.*

Proof. Use Lemma 2.5 in the closed-set lattice to obtain F . The lower inequality in (2.2) gives $\lambda + \|X \setminus F\| \leq \lambda + n$. Ordinary addition is strictly increasing in the right argument, so $\|X \setminus F\| \leq n$. By Lemma 2.6, the complement has some finite cardinality $m \leq n$. The upper inequality gives

$$\lambda + n \leq \lambda \oplus m = \lambda + m.$$

Hence $m = n$. □

This is stronger than merely finding a small residual space: after passing to T_0 , a finite ordinal tail is an actual finite set of points. Nothing here assumes that the specialization order is a wqo.

3 The closed-set structure of a Hoare powerspace

Definition 3.1. For a space X , the points of HX are its closed subsets, including $\emptyset$. Its *lower Vietoris topology* has subbasis

$$\diamond O = \{C \in C(X) : C \cap O \neq \emptyset\}, \quad O \text{ open in } X.$$

For a closed $F \subseteq X$, write

$$\square(F) = \{C \in C(X) : C \subseteq F\}.$$

The set $\square(F)$ is closed, being the complement of $\diamond(X \setminus F)$. Its points are themselves closed subsets of X ; this two-level notation should not be confused with F as a subspace of X .

Proposition 3.2. *If X is Noetherian, then HX is Noetherian, and its closed subsets are exactly*

$$D = \bigcup_{i=1}^k \square(F_i), \quad F_i \in C(X), \quad k < \omega. \quad (3.1)$$

The case $k = 0$ denotes the empty subset of the powerspace.

Proof. Finite unions in (3.1) form a family $\mathcal{D}$ closed under finite intersections, because $\square(F) \cap \square(G) = \square(F \cap G)$. They include the whole space. We first show that $\mathcal{D}$ is well-founded under strict inclusion.

Each nonempty member has a unique finite antichain of maximal generators: remove every F_i contained in another generator. For this irredundant set of generators $M(D)$ define

$$w(D) = \bigoplus_{F \in M(D)} \omega^{r_X(F)}, \quad w(\emptyset) = 0.$$

Suppose $D \subsetneq D'$. Each generator of D lies below a generator of D' . A generator of D not also a generator of D' must lie *strictly* below some generator of D' that is absent from $M(D)$; otherwise the maximality of the generators of D would fail. There is at least one such removed generator. Let γ be the largest rank of a generator removed from $M(D')$. Every newly added generator has rank below the rank of some removed generator, and hence below γ . At exponents above γ the two Cantor normal forms agree; at exponent γ the coefficient decreases. Thus $w(D) < w(D')$.

An arbitrary intersection of members of $\mathcal{D}$ is a finite intersection. Indeed, among all their finite intersections choose one minimal under inclusion. Intersecting it with any additional member cannot make it smaller, by minimality. It is therefore their full intersection. This proves that $\mathcal{D}$ is closed under arbitrary intersections. Since the closed subbasis consists of the sets $\square(F)$, it follows that $\mathcal{D}$ is precisely the closed-set family of HX . Its well-foundedness proves Noetherianity. □

Several consequences are useful. First, HX is T_0 , with specialization order inclusion: the closure of a point F is exactly $\square(F)$. Second, if F is closed in X , then

$$\square(F) \cong HF \quad (3.2)$$

as spaces. To check the topology, restrict each $\diamond O$ to the closed subsets of F and replace O by $O \cap F$. Third, any isomorphism $C(X) \cong C(Y)$ induces a homeomorphism $HX \cong HY$, since (3.1) depends only on that order. This justifies the earlier reduction to the T_0 quotient.

Lemma 3.3. *If $f : X \rightarrow Y$ is continuous and onto, the map*

$$f_* : HX \longrightarrow HY, \quad C \longmapsto \text{cl}_Y(f[C])$$

is continuous and onto. For a finite disjoint sum,

$$H(X_1 \sqcup \cdots \sqcup X_k) \cong HX_1 \times \cdots \times HX_k. \quad (3.3)$$

Proof. An open set meets $\text{cl}_Y(f[C])$ if and only if it meets $f[C]$, so $f_*^{-1}(\diamond O) = \diamond(f^{-1}(O))$. For a closed $B \subseteq Y$, choose $C = f^{-1}(B)$ to obtain $f_*(C) = B$.

For the disjoint-sum assertion, a closed set is a tuple of closed sets in the summands. Under this bijection a hit-open set becomes a union of coordinate hit-open sets. Conversely, a coordinate hit-open set is obtained by using an open subset contained in that summand. These observations give equality of the topologies. $\square$

Lemma 3.4 (Finite discrete extension). *Let D_m be a discrete space of m points. For every Noetherian F ,*

$$\|H(F \sqcup D_m)\| \leq \|HF\| \otimes 2^m.$$

Proof. There are 2^m points in HD_m . By (3.3), the powerspace in question is $HF \times HD_m$. Cover it by its 2^m slices, one for each point of the second factor. Each slice is homeomorphic to HF . The finite-union bound gives the natural sum of 2^m copies of $\|HF\|$, which is $\|HF\| \otimes 2^m$. The slices need not be closed; Corollary 2.4 deliberately allows this. $\square$

3.1 Discarding non-finitary points without changing stature

Let $H_{\text{fin}}X$ be the subspace of HX consisting of closures of finite subsets of X . A closure in this definition need not be finite as a set. The following lattice observation is particularly important later; it is the mechanism behind the equality also recorded in [1, Proposition 11.4].

Proposition 3.5. *For Noetherian X , taking traces on $H_{\text{fin}}X$ gives an isomorphism*

$$C(HX) \cong C(H_{\text{fin}}X).$$

In particular, $\|HX\| = \|H_{\text{fin}}X\|$.

Proof. Surjectivity is the definition of the subspace topology. To prove order reflection, suppose D, E are closed in HX and $D \not\subseteq E$. Choose a point $C \in D \setminus E$ and write $E = \bigcup_{i=1}^k \square(F_i)$. For each i choose $x_i \in C \setminus F_i$. The closed set $A = \text{cl}_X\{x_1, \dots, x_k\}$ is contained in C , so it lies in D , but it is contained in none of the F_i , so it lies outside E . It is a point of $H_{\text{fin}}X$. If $k = 0$, use $A = \emptyset$; every nonempty closed subset of HX contains this point. Thus the traces distinguish every failure of inclusion. An order isomorphism of closed lattices preserves top ranks. $\square$

4 Binary ordinal exponentiation

All occurrences of 2^α in this article mean ordinary *ordinal* exponentiation, not cardinal exponentiation and not a natural exponentiation. In particular $2^\omega = \omega$.

Lemma 4.1. *Write an ordinal uniquely as $\alpha = \omega \cdot q + n$, $n < \omega$. Then*

$$2^\alpha = \omega^q \cdot 2^n. \quad (4.1)$$

If λ is a nonzero limit ordinal, 2^λ is a limit ordinal and is indecomposable under finite natural sums of smaller ordinals. Moreover, for all ordinals α, β ,

$$2^{\alpha \oplus \beta} = 2^\alpha \otimes 2^\beta. \quad (4.2)$$

Proof. Ordinal exponent laws give

$$2^{\omega q + n} = (2^\omega)^q \cdot 2^n = \omega^q \cdot 2^n.$$

For a nonzero limit, $n = 0$ and $q > 0$. Every ordinal less than ω^q has only Cantor normal form exponents less than q ; a finite natural sum still has this property. This proves the indecomposability assertion.

For (4.2), left multiplication by ω changes each Cantor normal form exponent ξ to $1 + \xi$. This transformation is strictly increasing, and hence preserves the merging of coefficients used in natural sum. Thus, writing $\alpha = \omega q + n$ and $\beta = \omega r + m$,

$$\alpha \oplus \beta = \omega(q \oplus r) + (n + m).$$

The finite remainders cannot combine with the terms having positive exponent. Formula (4.1) now yields

$$2^{\alpha \oplus \beta} = \omega^{q \oplus r} 2^{n+m} = (\omega^q 2^n) \otimes (\omega^r 2^m).$$

□

The function $\alpha \mapsto 2^\alpha$ is strictly increasing and continuous at limit ordinals. A limit input turns its value into a single ω -monomial. That fact, rather than a coarse comparison with ω^α , is the key to the next induction.

5 The sharp universal upper bound

Theorem 5.1. *For every Noetherian space X ,*

$$1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}. \quad (5.1)$$

Proof. We may replace X by its T_0 quotient. Put $\alpha = \|X\|$ and prove the upper bound by transfinite induction on α .

Finite input. If $\alpha = n < \omega$, then X has exactly n points. The T_0 space HX is finite and has $|C(X)| \leq 2^n$ points. Its stature equals that cardinality, proving the bound. This includes $X = \emptyset$, whose powerspace is a singleton of stature $1 = 2^0$.

Nonzero limit input. Suppose α is a limit ordinal. Every proper closed $D \subsetneq HX$ has the form $\bigcup_{i=1}^k \square(F_i)$ with each $F_i \subsetneq X$. Let $\beta_i = \|F_i\| < \alpha$. By the finite-union bound, the principal powerspace homeomorphism, and the induction hypothesis,

$$\|D\| \leq \bigoplus_{i=1}^k \|HF_i\| \leq \bigoplus_{i=1}^k 2^{\beta_i} < 2^\alpha.$$

The final strict inequality is Lemma 4.1. The empty D satisfies the same inequality. Since 2^α is a limit ordinal, $\|D\| + 1 < 2^\alpha$. Taking the supremum over all proper closed D gives $\|HX\| \leq 2^\alpha$.

Infinite successor input. Write $\alpha = \lambda + n$, where λ is a nonzero limit and $0 < n < \omega$. By Corollary 2.8, choose a closed F of stature λ with an n -point complement. Replace the topology on this complement by the discrete topology and separate it from F . The resulting identity $F \sqcup D_n \rightarrow X$ is continuous and onto, so Lemmas 3.3 and 3.4 give

$$\|HX\| \leq \|H(F \sqcup D_n)\| \leq \|HF\| \otimes 2^n \leq 2^\lambda \otimes 2^n = 2^{\lambda+n}.$$

Here the penultimate step uses the induction hypothesis at λ ; the final identity follows from Lemma 4.1.

For the lower bound, $\square(\emptyset) = \{\emptyset\}$ has stature 1. Induction on $F \in C(X)$ gives

$$r_{HX}(\square(F)) \geq 1 + r_X(F).$$

Indeed, each $\square(E)$ with $E \subsetneq F$ is a proper closed subset of $\square(F)$, and ordinary left addition by 1 is continuous in its right argument. The base $F = \emptyset$ is the singleton just noted. Evaluate at $F = X$, for which $\square(X) = HX$. $\square$

The upper-bound argument does not use the general product formula for Noetherian spaces. Its only product estimate is the finite-slice cover in Lemma 3.4. In particular, the key extension beyond wqos is not hidden inside an unproved product assertion.

6 Sharpness at every ordinal, with T_1 upper witnesses

6.1 The lower endpoint

Proposition 6.1. *For every ordinal α , there is a Noetherian space L_α with*

$$\|L_\alpha\| = \alpha, \quad \|HL_\alpha\| = 1 + \alpha.$$

Proof. Give the well-order α its Alexandroff topology of upward-closed sets. Its closed sets are its initial segments, of lengths $\beta \leq \alpha$, so its stature is α .

The finitary closed subsets are the empty set and the principal initial segments $\downarrow x$, $x < \alpha$. Under inclusion they form a chain of type $1 + \alpha$. On this set of points the lower Vietoris topology is exactly the Alexandroff topology: the upper cone of $\downarrow x$ is the trace of $\diamond(\uparrow x)$, and the upper cone of the empty set is the whole space. An ordinal chain with this topology has stature its order type. Therefore $\|H_{\text{fin}}L_\alpha\| = 1 + \alpha$; use Proposition 3.5. $\square$

6.2 A finite-support bouquet

We now use one substantial external theorem:

$$\|Y \times Z\| = \|Y\| \otimes \|Z\| \quad (Y, Z \text{ Noetherian}). \quad (6.1)$$

This is the product theorem of Goubault-Larrecq and Labourel [1, Theorem 10.9]. The following all-ordinal T_1 sharpness construction depends on it; the preceding upper bound and the later spectrum classification do not.

Definition 6.2. For an infinite family $(X_i)_{i \in I}$ of nonempty Noetherian spaces, let $B(X_i : i \in I)$ have underlying set $\coprod_{i \in I} X_i$. Its closed subsets are the entire set and all sets

$$\coprod_{i \in S} F_i, \quad S \subseteq I \text{ finite}, \quad F_i \text{ closed in } X_i.$$

We call this a *finite-support bouquet*. No additional generic point is adjoined.

Lemma 6.3. *The stated family is the closed-set family of a Noetherian topology. Every finite union of whole components is closed and has the ordinary finite-disjoint-sum topology. If every X_i is T_1 , so is the bouquet.*

Proof. Finite unions preserve finite support. For an arbitrary intersection, there is nothing to check if every member is the whole space. Otherwise fix one finite-support member; the intersection is confined to those finitely many components and is closed in each of them. It is therefore again a permitted closed set.

Once a descending chain leaves the whole space, it lies in a finite union of components. This finite sum is Noetherian, so the chain stabilizes. The subspace-topology assertion follows directly by taking traces of the defining closed sets. Finally, every singleton is closed when each summand is T_1 . $\square$

The finite-support idea is related to the successor construction in [1, Example 11.9]. Omitting an added generic point keeps the present construction T_1 , and the limit-stage construction below works with an arbitrary ordinal-indexed family.

Theorem 6.4. *For every ordinal α there is a T_1 Noetherian space T_α such that*

$$\|T_\alpha\| = \alpha, \quad \|HT_\alpha\| = 2^\alpha.$$

When α is countable, the witness can be countable.

Proof. First construct, by recursion on ρ , a space B_ρ satisfying

$$\|B_\rho\| = \omega^\rho, \quad \|HB_\rho\| = 2^{\omega^\rho}. \quad (6.2)$$

Let B_0 be a singleton; its powerspace has two points, as required.

At a successor stage, let $B_{\rho+1}$ be the bouquet of countably many copies of B_ρ . The closed union of k whole components has stature $\omega^\rho \cdot k$. Every proper closed subset lies in such a finite union. It follows from the rank definition that

$$\|B_{\rho+1}\| = \sup_{k < \omega} (\omega^\rho \cdot k + 1) = \omega^{\rho+1}.$$

The powerspace of that closed k -component subspace embeds as a closed subspace of $HB_{\rho+1}$. By (3.3), the external product formula, and (4.2), its stature is

$$\bigotimes_{i=1}^k 2^{\omega^\rho} = 2^{\omega^\rho \cdot k}.$$

Taking the supremum over k gives the lower bound $2^{\omega^{\rho+1}}$, by continuity of exponentiation. The upper bound is theorem 5.1, so equality holds.

At a nonzero limit ρ , form the bouquet of one copy of B_ξ for each $\xi < \rho$. Every finite natural sum of the ordinals ω^ξ , $\xi < \rho$, is below ω^ρ , while the individual component statures are cofinal in ω^ρ . The rank definition therefore gives $\|B_\rho\| = \omega^\rho$. Each component is closed, so the stature of its powerspace is a lower bound on $\|HB_\rho\|$. Hence

$$\|HB_\rho\| \geq \sup_{\xi < \rho} 2^{\omega^\xi} = 2^{\omega^\rho}.$$

Again use the universal upper bound to get equality. This proves (6.2) at every ordinal stage.

For $\alpha > 0$, expand the finite coefficients in its Cantor normal form and write

$$\alpha = \omega^{\rho_1} + \cdots + \omega^{\rho_k}, \quad \rho_1 \geq \cdots \geq \rho_k.$$

Take $T_\alpha = B_{\rho_1} \sqcup \cdots \sqcup B_{\rho_k}$. Its stature is the natural sum of the summand statures, equal to the displayed Cantor normal form. Its Hoare powerspace is their product, whose stature is

$$\bigotimes_{i=1}^k 2^{\omega^{\rho_i}} = 2^{\bigoplus_{i=1}^k \omega^{\rho_i}} = 2^\alpha.$$

For $\alpha = 0$ take the empty space. Every construction preserves T_1 . Recursion over countable ordinals uses only countable bouquets and finite sums, proving the countability assertion. $\square$

Combining theorems 5.1 and 6.4 and Proposition 6.1 determines both extreme points of $\text{Spec}_H(\alpha)$ for every ordinal α .

7 Exactly when the older upper bound is attainable

Theorem 7.1. *For an infinite ordinal α , the following conditions are equivalent:*

- (i) *There is a Noetherian X with $\|X\| = \alpha$ and $\|HX\| = \omega^\alpha$.*
- (ii) $2^\alpha = \omega^\alpha$.
- (iii) $\omega \cdot \alpha = \alpha$.
- (iv) *Every exponent in the Cantor normal form of α is infinite.*
- (v) $\alpha = \omega^\omega \cdot \delta$ for some nonzero ordinal δ .

Proof. The sharp upper bound proves the equivalence of (i) and (ii), since $2^\alpha \leq \omega^\alpha$ and a witness for 2^α always exists. Write $\alpha = \omega q + n$ with $n < \omega$. Uniqueness of Cantor normal form in

$$2^\alpha = \omega^q 2^n$$

shows that this equals ω^α exactly when $n = 0$ and $q = \alpha$. These are precisely the conditions $\alpha = \omega\alpha$.

Write $\alpha = \omega^{\rho_1} c_1 + \dots + \omega^{\rho_k} c_k$ in Cantor normal form. Left multiplication by ω replaces ρ_i by $1 + \rho_i$. The normal forms agree exactly when every $1 + \rho_i = \rho_i$, that is, when every ρ_i is infinite.

Finally, multiplying a Cantor normal form on the left by ω^ω changes its exponents ξ to $\omega + \xi$. This is a strictly increasing map whose range consists of all ordinals at least ω : for any $\rho \geq \omega$, the order type of the tail after an initial segment of type ω supplies ξ with $\rho = \omega + \xi$. Thus precisely the normal forms in (iv) are obtained in (v). $\square$

Input stature α	Sharp maximum 2^α	Equals ω^α ?
ω	ω	No
$\omega + n, 0 < n < \omega$	$\omega \cdot 2^n$	No
ω^2	ω^ω	No
ω^3	$\omega \omega^2$	No
ω^ω	$\omega \omega^\omega$	Yes
$\omega^\omega + 1$	$\omega \omega^\omega \cdot 2$	No
$\omega^\omega + \omega$	$\omega \omega^{\omega+1}$	No
$\omega^{\omega+1}$	$\omega \omega^{\omega+1}$	Yes
ε_0	ε_0	Yes

Table 1: Consequences of the binary-exponent formula. In the last row, ε_0 is the least positive fixed point of $\alpha \mapsto \omega^\alpha$.

This settles the upper-attainment part of the chosen question, but extremal values alone do not classify the intervening spectrum. The next sections give such a classification at the first infinite ordinal and all of its finite successors.

8 A canonical infinite core at stature $\omega + n$

For a finite poset P , let $\mathcal{J}(P)$ be its poset of downsets, ordered by inclusion, and let $j(P) = |\mathcal{J}(P)|$. The empty downset is included, so $j(\emptyset) = 1$.

Proposition 8.1. *Let X be a T_0 Noetherian space with $\|X\| = \omega + n$, $n < \omega$. There is a unique closed subset F of stature ω . Its complement V has exactly n points. Every closed subset of X is either finite or contains F . The infinite closed subsets are exactly*

$$F \cup D, \quad D \in \mathcal{J}(Q), \quad (8.1)$$

where Q is the specialization poset on V .

Proof. For $n = 0$ take $F = X$. For $n > 0$ use Corollary 2.8. Every proper closed subset of F has stature less than ω , and hence is finite by Lemma 2.6. Also F is infinite.

If a closed $C \subseteq X$ does not contain F , then $C \cap F$ is a proper closed subset of F , hence finite. Its remaining points lie in the finite set V , so C is finite. Conversely, a closed set containing F is infinite. The interval $[F, X]$ is isomorphic to the closed-set lattice of V . Since V is finite and T_0 , its closed sets are exactly the downsets of Q , giving (8.1).

Any other closed subset of stature ω must be infinite and hence contain F . If it strictly contained F , its rank would exceed $r_X(F) = \omega$. This proves uniqueness. $\square$

Define the distinguished downset

$$U = \{v \in V : \text{cl}_X\{v\} \text{ is finite}\}. \quad (8.2)$$

It is a downset of Q , because $v' \leq v$ implies $\text{cl}_X\{v'\} \subseteq \text{cl}_X\{v\}$. If $v \notin U$, its closure is infinite and therefore contains all of F . Thus U distinguishes the finite-closure points from those whose closure absorbs the whole core.

8.1 The generic-point issue

A point $g \in F$ can have closure all of F , even though every proper closed subset of F is finite. Such a point would make F itself finitarily generated, changing the apparent list of points of $\mathbf{H}_{\text{fin}} X$. It must not be silently assumed absent.

Lemma 8.2 (Generic-point normalization). *There is at most one point $g \in F$ with $\text{cl}_X\{g\} = F$. If it exists, put $X' = X \setminus \{g\}$. The trace map $C(X) \rightarrow C(X')$ is an order isomorphism. Removing g preserves $\|X\|$, $\|HX\|$, Q , and U . After this removal, every point of the remaining core has finite closure.*

Proof. Two points with closure F would be indistinguishable, contrary to T_0 . Every finite closed set avoids g , since any closed set containing it contains the infinite set F . Each infinite closed set has the unique form $F \cup D$, and its trace is $(F \setminus \{g\}) \cup D$.

Taking these traces preserves and reflects inclusion. For two finite closed sets nothing changes. For two infinite ones inclusion is controlled by their D parts. A trace of an infinite set cannot be contained in a finite set, since $F \setminus \{g\}$ is still infinite. For a finite set contained in the trace of an infinite one, the same inclusion held before removal. The trace map is onto the closed sets of the subspace by definition, hence is an order isomorphism.

Stature is therefore preserved, and the powerspaces are homeomorphic by Proposition 3.2. The specialization order restricted to V is unchanged. An infinite closure loses at most g and stays infinite; a finite closure never contained g . Thus U is unchanged. Every other point of F already had a proper, therefore finite, closure. $\square$

For the rest of the spectrum argument, we perform this normalization and again call the resulting space X and its core F . This is a closed-lattice simplification, not a claim that the original space lacked a generic point.

9 A finite-cell rank lemma

The remaining transfinite computation can be packaged in an elementary lemma. It explains why a finite downset count becomes the coefficient of ω .

Lemma 9.1 (Finite-cell rank). *Suppose a T_0 space Y is partitioned into infinite sets A_p indexed by a finite poset P , and a finite exceptional set E . Assume:*

- (a) *Every closed $C \subseteq Y$ consists of all the cells indexed by some downset $I(C) \subseteq P$, together with finitely many other points.*
- (b) *For every downset $I \subseteq P$, the pure union $A_I = \bigcup_{p \in I} A_p$ is closed.*
- (c) *Every point belonging to a cell A_p has finite closure in Y .*

Then Y is Noetherian. For every closed C ,

$$r_Y(C) = \omega \cdot |I(C)| + |C \setminus A_{I(C)}|. \quad (9.1)$$

In particular,

$$\|Y\| = \omega \cdot |P| + |E|. \quad (9.2)$$

Proof. The set $I(C)$ is uniquely determined: a cell is infinite, so a finite remainder cannot contain an additional entire cell. Define $w(C)$ to be the right side of (9.1).

If $C \subseteq C'$, then $I(C) \subseteq I(C')$. If the index sets are different, the coefficient of ω strictly increases. If they agree, the finite remainder strictly increases in cardinality. Thus $w(C) < w(C')$ in either case. No infinite descending chain of closed sets is possible, and induction gives $r_Y(C) \leq w(C)$.

For the reverse inequality, induct on $k = |I(C)|$. When $k = 0$, C is a finite T_0 space, so its stature is its cardinality. Let $k > 0$ and first consider the pure union A_I , $|I| = k$. Choose a maximal $p \in I$ and put $I' = I \setminus \{p\}$, again a downset. For any positive integer N , choose N distinct points in the infinite cell A_p . The union K_N of their closures is finite by (c). It lies inside A_I , since A_I is closed. The closed set

$$B_N = A_{I'} \cup K_N$$

is a proper subset of A_I , has exactly $k - 1$ full cells, and has at least N points outside those cells. The induction hypothesis gives

$$r_Y(B_N) \geq \omega \cdot (k - 1) + N.$$

Taking the supremum over N in the rank recursion for A_I yields $r_Y(A_I) \geq \omega \cdot k$.

Now return to an arbitrary closed C with $I(C) = I$. The complement $C \setminus A_I$ has some finite cardinality m . Apply the lower closed–open inequality inside C , with closed part A_I . Its finite T_0 complement has stature m , so

$$r_Y(C) = \|C\| \geq \|A_I\| + m \geq \omega \cdot k + m = w(C).$$

This matches the upper bound. Taking $C = Y$ proves (9.2). $\square$

It is essential that cell points have finite closure and that every pure downset union is closed. Merely partitioning a space into finitely many infinite pieces would not determine its stature.

10 The complete spectrum at $\omega + n$

10.1 The formula for an arbitrary space

Theorem 10.1. *Let X be a Noetherian space of stature $\omega + n$, $n < \omega$. Pass to T_0 , let F be the core of Proposition 8.1, let Q be the n -point complement poset, and let U be (8.2). Then*

$$\|HX\| = \omega \cdot j(U) + (j(Q) - j(U)). \quad (10.1)$$

Proof. Use Lemma 8.2 to remove a generic core point if necessary. Every point of F , and every point of U , now has finite closure. A finite union of their closures is finite. A finite generating set has infinite closure precisely when it contains some point of $V \setminus U$.

It follows that the points of $H_{\text{fin}}X$ are all finite closed sets, together with precisely the following infinite closed sets:

$$E = \{F \cup D : D \in \mathcal{J}(Q), D \not\subseteq U\}. \quad (10.2)$$

For the converse implicit in this assertion, if D meets $V \setminus U$, then the closure of the finite set D is exactly $F \cup D$. Indeed, its trace on V is the downset D , and one of its point closures contains F . Every finite closed set is trivially finitarily generated by all of its own points.

For $J \in \mathcal{J}(U)$, define a cell

$$A_J = \{C \in C(X) : C \text{ is finite and } C \cap V = J\}. \quad (10.3)$$

A downset of U is also a downset of Q , because U itself is a downset. Every finite closed set belongs to exactly one such cell.

Each A_J is infinite. The finite closed set $B_J = \bigcup_{v \in J} \text{cl}_X\{v\}$ has trace J on V . There are infinitely many different finite closed sets contained in F : otherwise their finite union would contain every point of the infinite core, because each such point has finite closure. More directly, one can repeatedly choose a core point outside the current finite closed union and adjoin its closure. Union with B_J therefore produces infinitely many distinct members of A_J .

The closure in $H_{\text{fin}}X$ of a point $C \in A_J$ consists of finitarily closed subsets contained in the finite set C . It is finite. Thus condition (c) of Lemma 9.1 holds.

We next verify its closed-set conditions, with index poset $P = \mathcal{J}(U)$. The trace of $\square(C)$ is finite when C is finite. When $C = F \cup D$ is infinite, its trace contains exactly the entire cells A_J with $J \subseteq D \cap U$, and some of the finitely many exceptional points in (10.2). By Proposition 3.2, every closed subset of $H_{\text{fin}}X$ is a finite union of such traces. Its full-cell indices are therefore a downset of $\mathcal{J}(U)$; all other points form a finite remainder. This is condition (a).

Conversely, if I is a downset of $\mathcal{J}(U)$, take the union of the traces of $\square(F \cup J)$ over the maximal $J \in I$. Each J is contained in U , so this trace has no exceptional point and consists exactly of the cells with index contained in J . The union is precisely $\bigcup_{J \in I} A_J$. It is closed, proving condition (b). For $I = \emptyset$ use the empty union.

The finite-cell lemma now gives

$$\|H_{\text{fin}}X\| = \omega \cdot |\mathcal{J}(U)| + |E|.$$

The downsets of Q contained in U are exactly the downsets of U , so $|E| = j(Q) - j(U)$. Finally use Proposition 3.5. $\square$

10.2 Every finite pair has a countable realization

Theorem 10.2. *For every finite poset Q of size n and every downset $U \subseteq Q$, there is a countable T_0 Noetherian space $X(Q, U)$ with*

$$\|X(Q, U)\| = \omega + n, \quad \|HX(Q, U)\| = \omega \cdot j(U) + (j(Q) - j(U)).$$

Proof. Let C be a countably infinite set disjoint from Q . On the underlying set $C \sqcup Q$, declare the closed subsets to be exactly the following:

$$A \sqcup D, \quad A \subseteq C \text{ finite}, \quad D \in \mathcal{J}(U); \quad (10.4)$$

$$C \sqcup D, \quad D \in \mathcal{J}(Q). \quad (10.5)$$

The empty set and the whole space occur. Finite unions preserve these forms, since unions of downsets are downsets. For an arbitrary intersection, if every member has form (10.5), so does the intersection. Otherwise fix a finite-type member; the intersection has finite C part and a downset contained in U , hence has form (10.4). This is a topology.

A descending chain of infinite-type closed sets can change only its finite Q part, hence can have only finitely many strict steps before it either stops or enters a finite-type closed set. Thereafter it lies inside a finite set. Thus the topology is Noetherian.

The subspace C is closed and has the cofinite topology, of stature ω : its proper closed subsets are all finite sets. The complement is exactly the finite T_0 space Q , of stature n . Both sides of the closed–open inequality equal $\omega + n$, so $\|X(Q, U)\| = \omega + n$.

For $v \in U$, its closure is $\downarrow_Q v$, a finite set. For $v \notin U$, its closure is $C \sqcup \downarrow_Q v$. Points of C have singleton closures. These closures are pairwise distinct, so the space is T_0 . They also show that the distinguished finite-closure downset is exactly the specified U . Apply theorem 10.1. $\square$

Corollary 10.3 (Exact finite-tail spectrum). *For every $n < \omega$,*

$$\text{Spec}_H(\omega + n) = \{\omega \cdot j(U) + (j(Q) - j(U)) : |Q| = n, U \in \mathcal{J}(Q)\}.$$

Every value has a countable witness.

Corollary 10.4 (Rigidity in the T_1 case). *If X is T_1 and $\|X\| = \omega + n$, then*

$$\|HX\| = \omega \cdot 2^n.$$

In fact X is the disjoint sum of an infinite cofinite space and an n -point discrete space.

Proof. The finite complement of the core is closed because X is T_1 . Thus the core and its complement are both clopen. The core has all finite subsets closed and no other proper closed subsets, hence is cofinite. Its finite complement is discrete. Equivalently, in the spectrum formula Q is an antichain and $U = Q$, giving $j(U) = 2^n$ and zero finite remainder. $\square$

The source space can have a complicated family of finite point closures inside its core. At stature $\omega + n$, none of that complexity changes the answer: after the justified normalization, only the finite pair (Q, U) survives in the formula.

11 Examples, gaps, and exhaustive finite data

11.1 The first spectra

For finite input n , a T_0 space is an n -point poset, and its powerspace has one point per downset. Therefore the finite-input spectrum is simply

$$\text{Spec}_H(n) = \{j(Q) : |Q| = n\}.$$

For infinite input with finite tail, the first three spectra are

$$\text{Spec}_H(\omega) = \{\omega\},$$

$$\text{Spec}_H(\omega + 1) = \{\omega + 1, \omega \cdot 2\},$$

$$\text{Spec}_H(\omega + 2) = \{\omega + 2, \omega + 3, \omega \cdot 2 + 1, \omega \cdot 2 + 2, \omega \cdot 3, \omega \cdot 4\}.$$

For $n = 1$, Q has two downsets. Choosing $U = \emptyset$ gives $\omega + 1$; choosing $U = Q$ gives $\omega \cdot 2$. Thus, for example, $\omega + 2$ lies between the extreme values but is not itself attainable at input $\omega + 1$.

For $n = 2$, there are two poset types. A chain has $j(Q) = 3$ and possible $j(U) = 1, 2, 3$, giving $\omega + 2$, $\omega \cdot 2 + 1$, and $\omega \cdot 3$. An antichain has $j(Q) = 4$ and possible $j(U) = 1, 2, 4$, giving $\omega + 3$, $\omega \cdot 2 + 2$, and $\omega \cdot 4$.

For $n = 3$, the eighteen values can be presented without writing a long ordinal list: for each coefficient a in table 2, take $\omega \cdot a + b$ for each finite tail b in the second column.

Coefficient a	Permitted tails b at input $\omega + 3$
1	3, 4, 5, 7
2	2, 3, 4, 6
3	1, 2, 3
4	0, 1, 2, 4
5	0
6	0
8	0

Table 2: The complete spectrum at $\omega + 3$, encoded as pairs (a, b) .

Both endpoints of every finite-tail spectrum have elementary witnesses. Take Q to be a chain and $U = \emptyset$ to get $\omega + (j(Q) - 1) = \omega + n$. Take Q to be an antichain and $U = Q$ to get $\omega \cdot 2^n$. The intervening gaps are restrictions on finite poset downset counts, not failures of ordinal continuity.

11.2 Enumeration and certificates

The accompanying `verify_spectra.py` uses only the Python standard library. For each n , it enumerates all subsets of the pairs (i, j) with $0 \leq i < j < n$, retaining exactly the transitive relations. Every finite poset admits a linear extension, so relabeling along one puts it into this collection. This covers every isomorphism type, usually more than once. The counts are *not* counts of unlabeled posets and *not* counts of all labeled posets.

For each retained relation it enumerates all downsets U , counts $j(Q)$ and $j(U)$, and stores the pair $(j(U), j(Q) - j(U))$. A concrete relation and a distinguished downset are retained as a witness for each distinct pair.

n	Naturally labeled transitive relations	$ \text{Spec}_H(\omega + n) $
0	1	1
1	1	2
2	2	6
3	7	18
4	40	53
5	357	154
6	4824	431

Table 3: Exhaustive finite enumeration performed for this manuscript.

Across these runs the program examines 101,659 pairs consisting of a retained poset and one of its downsets. The archive contains the full spectra in `spectra.csv`, one witness per distinct value in `witnesses.json`, and the run summary in `verification_results.json`. The spectrum-size sequence 1, 2, 6, 18, 53, 154, 431 is reported as computed data, not asserted to be new to integer-sequence databases.

11.3 An independent finite-model identity

There is a useful check of the realization formula that does not use ordinal ranks. Replace the infinite cofinite core by an N -point antichain C_N . Keep the order on Q , put each point of C_N below each point of $Q \setminus U$, and add no further cross-comparabilities. Because U is a downset, $Q \setminus U$ is an upset and this is a partial order. Call it $P_{N,Q,U}$.

Proposition 11.1. *For $N \geq 1$,*

$$j(P_{N,Q,U}) = 2^N j(U) + (j(Q) - j(U)).$$

Proof. A downset whose Q part is contained in U has any of the 2^N possible core parts, and its Q part has $j(U)$ choices. A downset whose Q part is not contained in U must contain all of C_N , and it has $j(Q) - j(U)$ choices for its Q part. These cases are disjoint and exhaustive. $\square$

For every retained Q with at most four points, every downset U , and each $N = 1, 2, 3$, the program builds this finite poset and independently enumerates its downsets. All 1,221 checks matched the formula. This is an arithmetic and construction check, *not* a computer proof of theorem 10.1. In particular, taking an ordinary supremum of the finite counts would lose the ordinal coefficient information; the finite-cell lemma is what justifies the transfinite rank.

12 Dependency audit and the boundary of the result

The argument has two largely independent parts. The universal bound uses finite unions, the closed–open rank inequality, and the fact that 2^λ is naturally additively indecomposable at a nonzero limit λ . The finite-tail spectrum uses the unique core, the finitary closed-lattice isomorphism, and the finite-cell rank computation. Their overlap is mostly notation and elementary rank monotonicity.

Several potential errors deserve explicit attention.

Ordinal, not cardinal, arithmetic. The expression 2^ω here is ω . Also, the lower bound is $1 + \alpha$, not $\alpha + 1$. At $\alpha = \omega$ the former is ω , and replacing it by the latter would contradict the exact spectrum. Natural sums occur in finite disjoint sums and natural products in the imported product theorem; ordinary sums occur in a closed–open lower bound and in the final normal form $\omega a + b$.

The empty powerspace point. Even when X is empty, HX has the one point $\emptyset$. For a finite n -point T_0 space, the stature of HX is its number of closed subsets, not that number minus one. These conventions fix the base cases and the finite remainders in the spectrum formula.

No hidden assumption that all point closures are finite. At stature $\omega + n$, a core generic point may exist. It is removed only after proving an isomorphism of closed lattices. A point outside U is never removed: its infinite closure is exactly what produces one of the exceptional points in the finitary powerspace.

The product dependency is localized. Equation (6.1) is a published theorem, not reproved in this manuscript. It is used to certify the lower bounds for the T_1 bouquet witnesses. The universal upper proof, the old-bound nonattainment implications, and the complete $\omega + n$ classification can be read without it. Existing wqo sharpness witnesses give a separate route to unrestricted upper-bound attainment.

What has and has not been solved. The manuscript supplies a proof of a sharper general bound, witnesses for both extreme points at every ordinal, an exact answer to the upper-attainment subquestion, and the full spectrum for $\omega + n$. It does not classify $\text{Spec}_H(\omega \cdot 2)$, $\text{Spec}_H(\omega^2)$, or all higher spectra. There is no claim that a spectrum is an interval; the first nontrivial finite tails already show otherwise.

Verification status. The displayed transfinite arguments are mathematical proofs proposed here and have been checked against the stated definitions during preparation. They have not received independent human peer review and have not been formalized in Lean or another proof assistant. The included executable checks concern only finite posets and the finite realization identity. The novelty search located the decisive earlier wqo theorem and the original Noetherian question, but does not establish priority for the general extension or the spectrum theorem.

12.1 Specific next mathematical questions

The next natural target is stature $\omega \cdot 2$. The simple core argument no longer makes all proper closed subsets finite: some can already have stature ω , and their powerspace pieces have transfinite internal structure. A promising replacement for the finite-cell lemma would attach ordinal weights, rather than finite cardinalities, to a finite poset of closed strata. One would need precise hypotheses ensuring that finite unions of fragments cannot silently complete a stratum.

A second question is purely finite: describe the set of pairs $(j(U), j(Q) - j(U))$ for n -point Q without enumerating all posets. The spectrum theorem reduces the first infinite layer completely to that finite combinatorial problem. Bounds on possible downset counts, structural recurrences for ordinal sums and disjoint sums of finite posets, and efficient isomorph rejection could all improve the computation without touching the transfinite proof.

A third question concerns the T_1 subclass at higher statures. At $\omega + n$ it is rigid and always attains the maximum. The bouquet construction provides maximal T_1 examples everywhere, but it does not prove that every T_1 space at every ordinal stature is maximal. No such extrapolation is used in this article.

A Reproducing the artifacts

The archive contains this article as `.tex` and `.pdf`, a reusable BibTeX file, the Python program, computed spectra, explicit finite witnesses, and proof-audit and literature notes. Third-party source PDFs and font files are not included.

With Python 3.9 or later, run

```
python verify_spectra.py
```

The defaults cover $0 \leq n \leq 6$ and finite-model checks through $n = 4$. Options are `-max-n`, `-finite-check-n`, and `-output`; the input guard allows at most $n = 7$ because the enumeration is exponential. Details and file descriptions are in `README.md`.

With a normal TeX Live installation, compile using

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The bibliography is embedded in the TeX source, so a separate BibTeX run is unnecessary.

References

- [1] Jean Goubault-Larrecq and Bastien Laboureix. *Statures and sobrification ranks of Noetherian spaces*. Houston Journal of Mathematics **49**(1) (2023), 1–76. Public version consulted: arXiv:2112.06828v5, submitted 24 February 2023; 68-page preprint, with Question 1 on printed page 63. arXiv:2112.06828v5.
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